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CAPACITOR STRUCTURE AND METHOD FOR FABRICATING THE CAPACITOR

Abstract

A capacitor structure and a method for fabricating the capacitor are provided. The method includes forming a stack of a first electrode layer, a first dielectric layer, a second electrode layer, a second dielectric layer and a third electrode layer; forming a first spacer on the second dielectric layer and patterning the second dielectric layer and the second electrode layer by using the first spacer as a mask; forming a second spacer on the first dielectric layer and patterning the first dielectric layer and the first electrode layer by using the first spacer and the second spacer as a mask; and forming a first conductive layer on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer.

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Background/Summary

BACKGROUND

[0001] Integrated chips are formed on semiconductor die including millions or billions of transistor devices. The transistor devices are configured to act as switches and/or to produce power gains so as to enable logical functionality for an integrated chip (e.g., form a processor configured to perform logic functions). Integrated chips also include passive devices, such as capacitors, resistors, inductors, varactors, etc. Therefore, the improved capacitors and the improved process of fabricating the capacitors are desired as a development of a semiconductor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 to FIG. 32 illustrate the cross-sectional views of intermediate stages in the formation of a metal-insulator-metal (MIM) device in accordance with some embodiments.

[0004] FIG. 33 to FIG. 37 illustrate the cross-sectional views of intermediate stages in the formation of a MIM device in accordance with some embodiment.

DETAILED DESCRIPTION

[0005] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0006] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0007] A capacitor structure and a method for fabricating the capacitor are provided. At least one pair of self-aligned spacers are utilized in the fabrication method for forming the capacitor structure to decrease the number of photomasks used in patterning processes of material layers in the capacitor structure. Therefore, the fabrication costs of the capacitor structure can be lowered.

[0008] FIG. 1 to FIG. 32 illustrate the cross-sectional views of intermediate stages in the formation of a metal-insulator-metal (MIM) device in accordance with some embodiments.

[0009] Referring to FIG. 1, an interconnect structure including insulating layers 104, etch stop layers 105, interconnect wirings 106 and conductive vias 108 is formed over a substrate 102 (e.g., a

semiconductor substrate). The interconnect wirings **106** and the conductive vias **108** are alternatively stacked over the substrate **102** and are embedded in the insulating layers **104** and the etch stop layers **105**. The interconnect wirings **106** and the conductive vias **108** may be formed by, for example, CVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing. The insulating layers **104** and the etch stop layers **105** may be formed by, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), some other process, or a combination of the foregoing. While not shown in FIG. 1, it will be appreciated that one or more semiconductor devices (e.g., transistors, MOSFETs, etc.), may be formed in the substrate **102** by complementary metal-oxide-semiconductor (CMOS) processes.

[0010] A plurality of trenches **120** is formed on the interconnect wirings **106** such that the interconnect wirings **106** are revealed. The trenches **120** pass through the insulating layers **104** and the etch stop layers **105**, and reveal top surfaces of the interconnect wirings **106**. The trenches **120** are formed in the insulating layers **104** and the etch stop layers **105**. The trenches **120** extends from a top surface **104t** of the insulating layer **104** to the top surfaces of the interconnect wirings **106**. The trenches **120** are formed with substantially vertical sidewalls. In other embodiments, the trenches **120** are formed with inclined sidewalls. In some embodiments, at least one of the interconnect wirings **106** and at least one of the conductive vias **108** are disposed between two adjacent trenches **120**.

[0011] A plurality of conformal electrode layers **112a**, **116a**, **112b**, **116b** and a plurality of conformal dielectric layers **114a-114d** are formed in an alternating stack on the interconnect wirings **106** and lining the trenches **120**, and extends over the top surface **104t** of the insulating layer **104**. The electrode layer **112c1** is formed on the dielectric layer **114d** and laterally extends over the trenches **120** and the top surface **104t** of the insulating layer **104**. In some embodiments, as shown in FIG. 1, a first electrode layer **112a** extends over the top surface **104t** of the insulating layer **104** and lining the trenches **120**. The first electrode layer **112a** lands on and is electrically connected to the interconnect wirings **106**. A first dielectric layer **114a** conformally covers the first electrode layer **112a**. A second electrode layer **116a** conformally covers the first dielectric layer **114a**. A second dielectric layer **114b** conformally covers the second electrode layer **116a**. A third electrode layer **112b** conformally covers the second dielectric layer **114b**. A third dielectric layer **114c** conformally covers the third electrode layer **112b**. A fourth electrode layer **116b** conformally covers the third dielectric layer **114c**. A fourth dielectric layer **114d** conformally covers the fourth electrode layer **116b**. A fifth electrode layer **112c1** covers the fourth dielectric layer **114d**.

[0012] In some embodiments, the electrode layers **112a**, **116a**, **112b**, **116b**, **112c1** may be deposited by, for example, ALD, CVD, PVD, electrochemical plating, electroless plating, sputtering, some other deposition process, or a combination of the foregoing. The electrode layers **112a**, **116a**, **112b**, **116b**, **112c1** may be conductive and may be or include titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₅), aluminum (Al), aluminum titanium (AlTi), some other conductive material, or a combination of the foregoing. In some embodiments, the electrode layers **112a**, **116a**, **112b**, **116b**, **112c1** are deposited with a thickness between approximately 10 angstroms (Å) and approximately 1000 Å.

[0013] The dielectric layers **114a-114d** are formed between adjacent electrode layers of the electrode layers **112a**, **116a**, **112b**, **116b**, **112c1**. The dielectric layers **114a-114d** may be or include zirconium oxide (ZrO₂), aluminum oxide (Al₂O₃), hafnium oxide (HfO₂), hafnium aluminum oxide (HfAlO), tantalum oxide (Ta₂O₅), some other dielectric material, or any combination of the foregoing. In some embodiments, the dielectric layers **114a-114d** is or includes a metal oxide (e.g., ZrO₂, Al₂O₃, HfO₂, HfAlO, Ta₂O₅, or the like) and/or is or includes a high-k dielectric material. The dielectric layers **114a-114d** may be deposited or grown by ALD, CVD, PVD, thermal oxidation, some other deposition or growth process, or a combination of the foregoing. In some embodiments, the

dielectric layers **114a-114d** is formed with a thickness between about 10 Å and about 500 Å. In some embodiments, a number of the dielectric layer may be 1 or more, a number of the conductive may be 2 or more. The number of the dielectric layers and the number of the electrode layers are not limited thereto.

[0014] Then, an insulating layer **121a** is formed on the fifth electrode layer **112c1**. In some embodiments, the insulating layer **121a** may be formed by one or more deposition processes (e.g., a PVD process, a CVD process, a PECVD process, or the like). In various embodiments, the insulating layer **121a** may include silicon dioxide, silicon oxynitride, and/or the like.

[0015] Referring to FIG. 2, the insulating layer **121a** and the fifth electrode layer **112c1** are patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the insulating layer **121a** through the photolithography process, and then the insulating layer **121a** and the fifth electrode layer **112c1** are partially removed through the etch process until the fourth dielectric layer **114d** is revealed. An opening **O1** is formed to reveal the fourth dielectric layer **114d** and first sidewalls **S1** of the fifth electrode layer **112c1**. In some embodiments, the etching process may be performed by exposing the fifth electrode layer **112c1** to a first etchant. The first etchant may include a dry etchant (e.g., a reactive ion etching (RIE) etchant, a plasma etchant, or the like). In some embodiments, the first etchant may have an etching chemistry including one or more tetrafluoromethane (CF₄), fluoroform (CHF₃), chlorine (Cl₂), nitrogen (N₂), argon (Ar), boron trichloride (BCl₃), or the like. In some embodiments, after performing the removal process, any residual photoresist is removed by an ash process or by dissolution with a solvent.

[0016] Referring to FIG. 3, an additional insulating layer **121b** is deposited on the insulating layer **121a**, the first sidewalls **S1** of the fifth electrode layer **112c1** and the revealed fourth dielectric layer **114d**. In some embodiments, a nitride layer **124** are deposited on the additional insulating layer **121b**. In some embodiments, the additional insulating layer **121b** and the nitride layer **124** may be each formed by one or more deposition processes (e.g., a PVD process, a CVD process, a PECVD process, or the like). In various embodiments, the additional insulating layer **121b** may include silicon dioxide, silicon oxynitride, and/or the like. In various embodiments, the nitride layer **124** may include silicon nitride, silicon oxynitride, and/or the like. In some embodiments, a sum of a thickness of the additional insulating layer **121b** and the nitride layer **124** is in a range of between approximately 100 Å and approximately 1500 Å, between approximately 50 Å and approximately 1000 Å, between approximately 250 Å and approximately 7500 Å, approximately 500 Å, or other similar values.

[0017] Referring to FIG. 4, in a region **A1**, a first spacer **132** is formed on a first portion of the fourth dielectric layer **114d** to cover the first sidewall **S1** of the fifth electrode layer **112c1**. Then, the fourth dielectric layer **114d** and the fourth electrode layer **116b** are patterned by using the fifth electrode layer **112c1** and the first spacer **132** as a mask, to reveal the third dielectric layer **114c**, a first sidewall **S2** of the fourth dielectric layer **114d** and a first sidewall **S3** of the fourth electrode layer **116b**.

[0018] In a region **A2**, another first spacer **132** is formed in the same process and at same time as the first spacer **132**. In region **A2**, the first spacer **132** is formed on a second portion of the fourth dielectric layer **114d** to cover the first sidewall **S1** of the fifth electrode layer **112c1**. The region **A1** and the region **A2** are mirror-symmetrical to each other with respect to a center line (as an axis of symmetry) (not shown) of the opening **O1**. In order to described concisely and clearly, the following description only indicates the example structures and the example processes in the region **A1**.

[0019] The insulating layer **121a**, the additional insulating layer **121b** and the nitride layer **124** are removed by an etching process including a second etchant. The second etchant removes the insulating layer **121a**, the additional insulating layer **121b** and the nitride layer **124** horizontally

extending over the fifth electrode **112c1** and the fourth dielectric layer **114d**, to form the first spacer **132** covering the first sidewall **S1** of the fifth electrode layer **112c1** and reveals the fourth electrode layer **114d**. In some embodiments, the second etchant may include a dry etchant (e.g., a reactive ion etching (RIE) etchant, a plasma etchant, or the like). In some embodiments, the second etchant may have an etching chemistry including one or more tetrafluoromethane (CF_4), fluoroform (CHF_3), chlorine (Cl_2), nitrogen (N_2), argon (Ar), boron trichloride (BCl_3), or the like.

[0020] In some embodiments, portions of the fourth dielectric layer **114d** and the fourth electrode layer **116b** are subsequently removed by an etching process including a third etchant. Then, the third dielectric layer **114c** is revealed. In some embodiments, the third etchant may be a same etchant as the second etchant (e.g., as part of a continuous etching process that defines the first spacer **132**), while in other embodiments the third etchant may include an etchant that is separate and different than the second etchant. Since the fourth dielectric layer **114d** and the fourth electrode layer **116b** are etched according to the first spacer **132**, the first spacer **132** has an outermost sidewall that is substantially aligned with the first sidewall **S2** of the fourth dielectric layer **114d** and the first sidewall **S3** of the fourth electrode layer **116b**.

[0021] In some embodiments, a thickness **T1** of the fifth electrode layer **112c1** substantially equals to a height **H1** of the first spacer **132**. A first lateral offset **D1** is between the first sidewall **S3** of the fourth electrode layer **116b** and the first sidewall **S1** of the fifth electrode layer **112c1**, and the first lateral offset **D1** substantially equals to a maximum width **W1** of the first spacer **132**. In some embodiments, the first spacer **132** is disposed over a lateral offset region of the fourth electrode layer **116b**.

[0022] In some embodiments, using the first spacer **132** as the mask to define the fourth electrode layer **116b** allows patterning of the fourth dielectric layer **114d** and the fourth electrode layer **116b** without using a photomask, thereby providing for a relatively low-cost process to form a metal-insulator-metal (MIM) capacitor structure. Moreover, having the first spacer **132** in place during the etching process allows for the first spacer **132** to cover the first sidewall **S1** of the fifth electrode layer **112c1** and to prevent conductive byproducts (e.g., from etching the fourth electrode layer **116b**) from being re-deposited along the first sidewall **S1** of the fifth electrode layer **112c1**. By preventing conductive byproducts from being re-deposited along the first sidewall **S1** of the fifth electrode layer **112c1**, the conductive byproducts cannot form a conductive path between the fourth electrode layer **116b** and the fifth electrode layer **112c1**, thereby preventing electrical short between the fourth electrode layer **116b** and the fifth electrode layer **112c1**.

[0023] Referring to FIG. 5, unless further description as follows, the definition of the reference symbols and labeled representations are the same as FIG. 3, and will not be repeated herein. An insulating layer **122** is deposited on the patterned fifth electrode layer **112c1**, the first spacer **132**, the first sidewall **S2** of the fourth dielectric layer **114d**, the first sidewall **S3** of the fourth electrode layer **116b** and the revealed third dielectric layer **114c**. In some embodiments, a nitride layer **124** are deposited on the insulating layer **122**.

[0024] Referring to FIG. 6, a second spacer **134** is formed on a first portion of the third dielectric layer **114c** to cover the first spacer **132**, the first sidewall **S2** of the fourth dielectric layer **114d** and the first sidewall **S3** of the fourth electrode layer **116b**. Then, the third dielectric layer **114c** and the third electrode layer **112b** are patterned by using the fifth electrode layer **112c1**, the first spacer **132** and the second spacer **134** as a mask, to reveal the second dielectric layer **114b**, a first sidewall **S4** of the third dielectric layer **114c** and a first sidewall **S5** of the third electrode layer **112b**.

[0025] The insulating layer **122** and the nitride layer **124** are performed an etching process. to remove the insulating layer **122** and the nitride layer **124** horizontally extending over the fifth electrode **112c1** and the third dielectric layer **114c**. Then, the second spacer **134** is formed to cover the first sidewall **S2** of the fourth dielectric layer **114d** and the first sidewall **S3** of the fourth electrode layer **116b**. In some embodiments, the etching process may be performed as the same as

the etching process including the second etchant, and will not be repeated herein.

[0026] In some embodiments, portions of the third dielectric layer **114c** and the third electrode layer **112b** are subsequently performed an etching process. Then, the second dielectric layer **114b** is revealed. In some embodiments, the etching process may be performed as the same as the etching process including the third etchant, and will not be repeated herein. Since the third dielectric layer **114c** and the third electrode layer **112b** are etched according to the second spacer **134**, the second spacer **134** has an outermost sidewall that is substantially aligned with the first sidewall S4 of the third dielectric layer **114c** and the first sidewall S5 of the third electrode layer **112b**.

[0027] In some embodiments, the height H1 of the first spacer **132** is less than a height H2 of the second spacer **134**. A sum of a thickness T3 of the fourth electrode layer **116b**, a thickness T2 of the fourth dielectric layer **114d** and the thickness T1 of the fifth electrode layer **112c1** substantially equals to the height H2 of the second spacer **134**. A second lateral offset D2 is between the first sidewall S5 of the third electrode layer **112b** and the first sidewall S3 of the fourth electrode layer **116b**, and the second lateral offset D2 substantially equals to a maximum width W2 of the second spacer **134**. In some embodiments, the maximum width W2 of the second spacer **134** is greater than the maximum width W1 of the first spacer **132**. In some embodiments, the maximum width W2 of the second spacer **134** equals to the maximum width W1 of the first spacer **132**. In some embodiments, the maximum width W2 of the second spacer **134** is less than the maximum width W1 of the first spacer **132**. In some embodiments, the second spacer **134** is disposed over a lateral offset region of the third electrode layer **112b**. In some embodiments, a top surface **112c1t** of the fifth electrode layer **112c1** substantially levels to a top surface of the first spacer **132** and a top surface of the second spacer **134**.

[0028] In some embodiments, the second spacer **134** has the same effects as the first spacer **132**. Using the second spacer **134** as the mask to define the third electrode layer **112b** allows for removing the third dielectric layer **114c** and the third electrode layer **112b** without a photomask, thereby providing for a relatively low cost process to form the MIM capacitor structure and preventing electrical shorting between the fourth electrode layer **116b** and the third electrode layer **112b**.

[0029] Referring FIG. 7 and FIG. 8, a first conductive layer **112c2** is formed on the fifth electrode layer **112c1**, the first spacer **132**, and the second spacer **134**, and the first sidewall S5 of the third electrode layer **112b** to electrically connect the fifth electrode layer **112c1** and the third electrode layer **112b**.

[0030] In FIG. 7, the first conductive layer **112c2** is formed on the fifth electrode layer **112c1**, the first spacer **132**, and the second spacer **134**, the first sidewall S4 of the third dielectric layer **114c**, the first sidewall S5 of the third electrode layer **112b** and the revealed second dielectric layer **114b**. The first conductive layer **112c2** may be deposited by, for example, ALD, CVD, PVD, electrochemical plating, electroless plating, sputtering, some other deposition process, or a combination of the foregoing. The first conductive layer **112c2** may be conductive and may be or include, for example, titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₅), aluminum (Al), aluminum titanium (AlTi), some other conductive material, or a combination of the foregoing. In some embodiments, the first conductive layer **112c2** is deposited with a thickness between approximately 10 angstroms (Å) and approximately 1000 Å.

[0031] In FIG. 8, the first conductive layer **112c2** is patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the first conductive layer **112c2** through the photolithography process, and then the first conductive layer **112c2** is partially removed through the etch process until the second dielectric layer **114b** is revealed. In some embodiments, the etching process may be performed by exposing the first conductive layer **112c2** to a fourth etchant. The fourth etchant may include a dry etchant (e.g., a reactive ion etching (RIE) etchant, a plasma etchant, or the like). In some embodiments, the fourth etchant may have an

etching chemistry including one or more tetrafluoromethane (CF.sub.4), fluoroform (CHF.sub.3), chlorine (Cl.sub.2), nitrogen (N.sub.2), argon (Ar), boron trichloride (BCl.sub.3), or the like. In some embodiments, after performing the removal process, any residual photoresist is removed by an ash process or by dissolution with a solvent.

[0032] The first conductive layer **112c2** is formed on the fifth electrode layer **112c1**, the first spacer **132**, and the second spacer **134**, the first sidewall **S4** of the third dielectric layer **114c**, the first sidewall **S5** of the third electrode layer **112b** to electrically connect the fifth electrode layer **112c1** and the third electrode layer **112b**. The first conductive layer **112c2** is in contact with the top surface **112c1t** of the fifth electrode layer **112c1** and the first sidewall **S5** of the third electrode layer **112b**. The first conductive layer **112c2** is spaced apart from the fourth electrode layer **116b** by the first spacer **132** and the second spacer **134**.

[0033] Referring to FIG. 9 to FIG. 12, a third spacer **136** may be formed by the same processes and the same materials as the first spacer **132**. Unless further description as follows, the processes, the materials and the effects of forming the third spacer **136** are the same as the first spacer **132** of FIG. 3 to FIG. 4, and will not be repeated herein. A fourth spacer **138** may be formed by the same processes and the same materials as the second spacer **134**. Unless further description as follows, the processes, the materials and the effects of forming the fourth spacer **138** are the same as the second spacer **134** of FIG. 5 to FIG. 6, and will not be repeated herein.

[0034] The third spacer **136** is formed on a first portion of the second dielectric layer **114b** to cover a portion of the first conductive layer **112c2**. Then, the second dielectric layer **114b** and the second electrode layer **116a** are patterned by using the first conductive layer **112c2** and the third spacer **136** as a mask, to reveal the first dielectric layer **114a**, a first sidewall **S6** of the second dielectric layer **114b** and a first sidewall **S7** of the second electrode layer **116a**.

[0035] Since the second dielectric layer **114b** and the second electrode layer **116a** are etched according to the third spacer **136**, the third spacer **136** has an outermost sidewall that is substantially aligned with the first sidewall **S6** of the second dielectric layer **114b** and a first sidewall **S7** of the second electrode layer **116a**.

[0036] The fourth spacer **138** is formed on a first portion of the first dielectric layer **114a** to cover the third spacer **136**, the first sidewall **S6** of the second dielectric layer **114b** and a first sidewall **S7** of the second electrode layer **116a**. Then, the first dielectric layer **114a** and the first electrode layer **112a** are patterned by using the first conductive layer **112c2**, the third spacer **136** and the fourth spacer **138** as a mask, to reveal the insulating layer **104**, a first sidewall **S8** of the first dielectric layer **114a** and a first sidewall **S9** of the first electrode layer **112a**.

[0037] Since the first dielectric layer **114a** and the first electrode layer **112a** are etched according to the fourth spacer **138**, the fourth spacer **138** has an outermost sidewall that is substantially aligned with the first sidewall **S8** of the first dielectric layer **114a** and the first sidewall **S9** of the first electrode layer **112a**.

[0038] In some embodiments, a height **H3** of the third spacer **136** is less than a height **H4** of the fourth spacer **138**. The height **H2** of the second spacer **134** is less than the height **H3** of the third spacer **136**. The height **H1** of the first spacer **132** is less than the height **H2** of the second spacer **134**.

[0039] A third lateral offset **D3** is between the first sidewall **S5** of the third electrode layer **112b** and the first sidewall **S7** of the second electrode layer **116a**, and the third lateral offset **D3** substantially equals to a maximum width **W3** of the third spacer **136**. In some embodiments, the third spacer **136** is disposed over a lateral offset region of the second electrode layer **116a**. A fourth lateral offset **D4** is between the first sidewall **S9** of the first electrode layer **112a** and the first sidewall **S7** of the second electrode layer **116a**, and the fourth lateral offset substantially equals to a maximum width **W4** of the fourth spacer **138**. In some embodiments, the fourth spacer **138** is disposed over a lateral offset region of the first electrode layer **112a**.

[0040] In some embodiments, a top surface **112c2t** of the first conductive layer **112c2** substantially

levels to a top surface of the third spacer **136** and a top surface of the fourth spacer **138**. The different between the top surfaces of the spacers (the third spacer **136**, the fourth spacer **138** and the top surfaces of the spacers (the first spacer **132**, the second spacer **134** substantially equals to a thickness of the first conductive layer **112c2**.

[0041] Referring FIG. **13** and FIG. **14**, a second conductive layer **112c3** is formed on the top surface **112c2t** of the first conductive layer **112c2**, the third spacer **136**, and the fourth spacer **138**, and the first sidewall **S9** of the first electrode layer **112a** to electrically connect the fifth electrode layer **112c1** and the first electrode layer **112a**, and to electrically connect the third electrode layer **112b** and the first electrode layer **112a**.

[0042] In FIG. **13**, the second conductive layer **112c3** is formed on the first conductive layer **112c2**, the third spacer **136**, the fourth spacer **138**, the first sidewall **S8** of the first dielectric layer **114a**, the first sidewall **S9** of the first electrode layer **112a** and the revealed insulating layer **104**. The second conductive layer **112c3** may be deposited by the same processes and the same materials as the first conductive layer **112c2**.

[0043] In FIG. **14**, the second conductive layer **112c3** is patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the second conductive layer **112c3** through the photolithography process, and then the second conductive layer **112c3** is partially removed through the etch process until the insulating layer **104** is revealed. The second conductive layer **112c3** is in contact with the top surface **112c2t** of first conductive layer **112c2** and the first sidewall **S9** of the first electrode layer **112a**. The second conductive layer **112c3** is spaced apart from the second electrode layer **116a** by the third spacer **136** and the fourth spacer **138**.

[0044] Referring to FIG. **15**, an insulating material is deposited on the second conductive layer **112c3** and fills up the opening **O1**. The insulating material may be any suitable insulating materials. Any suitable planarization process (such as Chemical-Mechanical Polishing (CMP)) may be performed on the insulating material until reveal the second conductive layer **112c3**. Then, an insulating layer **104** is formed within opening **O1**. In some embodiments, a top surface of the second conductive layer **112c3** substantially levels to a top surface of the insulating layer **104**.

[0045] Referring to FIG. **16**, a fifth dielectric layer **114e** is formed on the second conductive layer **112c3** and the insulating layer **104** by the same processes and the same materials as the dielectric layers **114a-114d**.

[0046] Referring to FIG. **17**, the fifth dielectric layer **114e**, the second conductive layer **112c3**, the first conductive layer **112c2** and the fifth electrode layer **112c1** are patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the fifth dielectric layer **114e** through the photolithography process, and then the fifth dielectric layer **114e**, the second conductive layer **112c3**, the first conductive layer **112c2** and the fifth electrode layer **112c1** are partially removed through the etch process until the fourth dielectric layer **114d** is revealed. An opening **O2** is formed to reveal the fourth dielectric layer **114d**, second sidewall(s) **S11** of the fifth dielectric layer **114e**, second sidewall(s) **S12** of the second conductive layer **112c3**, sidewalls **S13** of the first conductive layer **112c2**, and sidewall(s) **S14** of the fifth electrode layer **112c1** (shown in FIG. **19**).

[0047] Referring to FIGS. **18** to **19**, a fifth spacer **142** in a region **B1** may be formed by the same processes and the same materials as the first spacer **132** after depositing the insulating layer **122** and the nitride layer **124**. Unless further description as follows, the processes, the materials and the effects of forming the fifth spacer **142** are the same as the first spacer **132** of FIG. **3** to FIG. **4**, and will not be repeated herein.

[0048] In a region **B1**, the fifth spacer **142** is formed on a third portion of the fourth dielectric layer **114d** to cover the second sidewall **S11** of the fifth dielectric layer **114e**, the second sidewall **S12** of the second conductive layer **112c3**, the second sidewall **S13** of the first conductive layer **112c2**, and

the second sidewall **S14** of the fifth electrode layer **112c1**. Then, the fourth dielectric layer **114d** and the fourth electrode layer **116b** are patterned by using the fifth dielectric layer **114e** and the fifth spacer **142** as a mask, to reveal the third dielectric layer **114c**, a second sidewall **S15** of the fourth dielectric layer **114d** and a second sidewall **S16** of the fourth electrode layer **116b**.

[0049] In a region **B2**, an additional fifth spacer **142** is formed in the same process and at same time as the fifth spacer **142**. In region **A2**, the additional fifth spacer **142** is formed on a fourth portion of the fourth dielectric layer **114d** to cover the second sidewall **S11** of the fifth dielectric layer **114e**, the second sidewall **S12** of the second conductive layer **112c3**, the second sidewall **S13** of the first conductive layer **112c2**, and the second sidewall **S14** of the fifth electrode layer **112c1**. The region **B1** and the region **B2** are mirror-symmetrical to each other with respect to a center line (as an axis of symmetry) (not shown) of the opening **02**. In order to described concisely and clearly, the following description only indicates the example structures and the example processes in the region **B1**.

[0050] Since the fourth dielectric layer **114d** and the fourth electrode layer **116b** are etched according to the fifth spacer **142**, the fifth spacer **142** has an outermost sidewall that is substantially aligned with the second sidewall **S15** of the fourth dielectric layer **114d** and the second sidewall **S16** of the fourth electrode layer **116b**.

[0051] In some embodiments, a height **H5** of the fifth spacer **142** substantially equals to a sum of a thickness **T1** of the fifth electrode layer **112c1**, a thickness of the first conductive layer **112c2**, a thickness of the second conductive layer **112c3**, and a thickness of the fifth dielectric layer **114e**. In some embodiments, a height **H5** of the fifth spacer **142** is greater than the height **H1** of the first spacer **132**. A fifth lateral offset **D5** is between the second sidewall **S16** of the fourth electrode layer **116b** and the second sidewall **S14** of the fifth electrode layer **112c1**, and the fifth lateral offset **D5** substantially equals to a maximum width **W5** of the fifth spacer **142**. In some embodiments, the fifth spacer **142** is disposed over a lateral offset region of the fourth electrode layer **116b**.

[0052] Referring to FIGS. **20** to **21**, a third conductive layer **116c1** is formed on the fifth dielectric layer **114e**, the fifth spacer **142** and the second sidewall **S16** of the fourth electrode layer **116b** to be in contact with the second sidewall **S16** of the fourth electrode layer **116b**. The third conductive layer **116c1** is electrically connected to the fourth electrode layer **116b**. Unless further description as follows, the processes and the materials of forming the third conductive layer **116c1** are the same as the first conductive layer **112c2** of FIG. **7** to FIG. **8**, and will not be repeated herein.

[0053] The third conductive layer **116c1** is patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the third conductive layer **116c1** through the photolithography process, and then the third conductive layer **116c1** is partially removed through the etch process until the third dielectric layer **114c** is revealed.

[0054] The third conductive layer **116c1** is formed on the fifth dielectric layer **114e**, the fifth spacer **142**, the second sidewall **S15** of the fourth dielectric layer **114d**, the second sidewall **S16** of the fourth electrode layer **116b** and revealed the third dielectric layer **114c**. The third conductive layer **116c1** is spaced apart from the fifth electrode layer **112c1** by the fifth spacer **142**.

[0055] Referring to FIGS. **22** to **23**, a spacer structure **143** may be formed by the same processes and the materials as the third spacer **136** after depositing the insulating layer **122** and the nitride layer **124**. Unless further description as follows, the processes, the materials and the effects as of forming the spacer structure **143** are the same as the third spacer **136** of FIG. **9** to FIG. **10**, and will not be repeated herein.

[0056] The spacer structure **143** is formed on a second portion of the third dielectric layer **114c** to cover a portion of the third conductive layer **116c1**. Then, the third dielectric layer **114c** and the third electrode layer **112b** are patterned by using the third conductive layer **116c1** and the spacer structure **143** as a mask, to reveal the second dielectric layer **114b**, a second sidewall **S17** of the third dielectric layer **114c** and a second sidewall **S18** of the third electrode layer **112b** (shown in

FIG. 19). In some embodiments, a top surface **116c1t** of the third conductive layer **116c1** substantially levels to a top surface **143t** of the spacer structure **143**.

[0057] Since the third dielectric layer **114c** and the third electrode layer **112b** are etched according to the spacer structure **143**, the spacer structure **143** has an outermost sidewall that is substantially aligned with the second sidewall **S17** of the third dielectric layer **114c** and the second sidewall **S18** of the third electrode layer **112b**.

[0058] Referring to FIGS. 24 to 25, a sixth spacer **144** is formed by removing a portion of the spacer structure **143** and a seventh spacer **146** is formed during a same process. The seventh spacer **146** may be formed by the same processes and the materials as the fourth spacer **138** after depositing the insulating layer **122** and the nitride layer **124**. Unless further description as follows, the processes, the materials and the effects of forming the seventh spacer **146** are the same as the fourth spacer **138** of FIG. 11 to FIG. 12, and will not be repeated herein.

[0059] After performing an etching process including the second etchant, the insulating layer **122** and the nitride layer **124** horizontally extending over the third conductive layer **116c1** and the second dielectric layer **114b** is removed. Then, the etching process also removes the third conductive layer **116c1** horizontally extending over the fifth spacer **142** and the fifth dielectric layer **114e**, and removes a portion of the spacer structure **143** and portions of the insulating layer **122** and the nitride layer **124**. The sixth spacer **144** and the seventh spacer **146** are formed.

[0060] A height **H6** of the sixth spacer **144** is less than a height of the spacer structure **143**. Except for the height, the property and the arrangement of the sixth spacer **144** are the same as the spacer structure **143**, and will not be repeated herein.

[0061] The seventh spacer **146** is formed on a second portion of the second dielectric layer **114b** to cover the sixth spacer **144**, the second sidewall **S17** of the third dielectric layer **114c** and the second sidewall **S18** of the third electrode layer **112b**. Then, the second dielectric layer **114b** and the second electrode layer **116a** are patterned by using the fifth dielectric layer **114e**, the fifth spacer **142**, the third conductive layer **116c1**, the sixth spacer **144** and the seventh spacer **146** as a mask, to reveal the first dielectric layer **114a**, a second sidewall **S19** of the second dielectric layer **114b** and a second sidewall **S20** of the second electrode layer **116a** (shown in FIG. 25).

[0062] Since the second dielectric layer **114b** and the second electrode layer **116a** are etched according to the seventh spacer **146**, the seventh spacer **146** has an outermost sidewall that is substantially aligned with the second sidewall **S19** of the second dielectric layer **114b** and the second sidewall **S20** of the second electrode layer **116a**.

[0063] The seventh spacer **146** substantially levels to the third conductive layer **116c1**, the sixth spacer **144**, and the fifth spacer **142**. A height **H7** of the seventh spacer **146** is greater than the height **H6** of the sixth spacer **144**. The height **H6** of the sixth spacer **144** is greater than the height **H5** of the fifth spacer **142**. A sixth lateral offset **D6** is between the second sidewall **S16** of the fourth electrode layer **116b** and the second sidewall **S18** of the third electrode layer **112b**, and the sixth lateral offset **D6** substantially equals to a maximum width **W6** of the sixth spacer **144**. In some embodiments, the sixth spacer **144** is disposed over a lateral offset region of the third electrode layer **112b**. A seventh lateral offset **D7** is between the second sidewall **S20** of the second electrode layer **116a** and the second sidewall **S18** of the third electrode layer **112b**, and the seventh lateral offset **D7** substantially equals to a maximum width **W7** of the seventh spacer **146**. In some embodiments, the seventh spacer **146** is disposed over a lateral offset region of the second electrode layer **116a**.

[0064] Referring FIG. 26, a sixth electrode layer **116c2** may be deposited by the same processes as the second conductive layer **112c3** of FIG. 13. In some embodiments, the sixth electrode layer **116c2** may be formed by the same materials as the fourth electrode layer **116b** and the second conductive layer **112c3**.

[0065] The sixth electrode layer **116c2** is formed on the fifth dielectric layer **114e**, the fifth spacer **142**, the third conductive layer **116c1**, the sixth spacer **144**, the seventh spacer **146**, the second

sidewall **S19** of the second dielectric layer **114b**, the second sidewall **S20** of the second electrode layer **116a** and the first dielectric layer **114a**, to electrically connect the sixth electrode layer **116c2** and the fourth electrode layer **116b**, and to electrically connect sixth electrode layer **116c2** and the second sidewall **S20** of the second electrode layer **116a**. The sixth electrode layer **116c2** is spaced apart from the third electrode layer **112b** by the sixth spacer **144** and the seventh spacer **146**.

[0066] Referring FIG. **27**, a capping layer **150** is formed on the sixth electrode layer **116c2**. The capping layer **150** is formed by, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), some other process, or a combination of the foregoing. In some embodiments, the capping layer **150** may include a dielectric material such as silicon oxynitride, silicon oxycarbide, or the like.

[0067] Referring FIG. **28** and FIG. **29**, any suitable photomasks may be used in the photolithography process to reveal the insulating layer **104** in the opening **O2** and the insulating layer **104** in an opening **O3**.

[0068] In FIG. **28**, the capping layer **150** and the sixth electrode layer **116c2** are patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the capping layer **150** through the photolithography process, and then the capping layer **150** and the sixth electrode layer **116c2** are partially removed through the etch process until the first dielectric layer **114a** is revealed in the opening **O2** and opening **O3**.

[0069] In FIG. **29**, the first dielectric layer **114a** and the first electrode layer **112a** are patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the first dielectric layer **114a** through the photolithography process, and then the first dielectric layer **114a** and the first electrode layer **112a** are partially removed through the etch process until the insulating layer **104** is revealed in the opening **O2** and opening **O3**. Recesses of the insulating layer **104** are formed within the insulating layer **104** in the opening **O2** and opening **O3**. Then, MIM capacitor structures **100** are formed. A single MIM capacitor structure **100** is an asymmetrical structure. The adjacent MIM capacitor structures **100** may be mirror-symmetrical to each other.

[0070] Referring FIG. **30**, an insulating material is deposited on the capping layer **150** and fills up the opening **O2** and the opening **O3** to form the insulating layer **104**. The insulating material may be any suitable insulating materials. Any suitable planarization process (such as Chemical-Mechanical Polishing (CMP)) may be performed on the insulating material. Then, the etch stop layers **105** and the insulating layer **104** may be alternatively formed on the insulating layer **104** until to a desired stack.

[0071] Referring FIG. **31**, the insulating layer **104** and the etch stop layers **105** are patterned by a photolithography process followed by an etch process, and a photomask is used in the photolithography process. In other words, a patterned photoresist layer (not shown) may be formed on the insulating layer **104** through the photolithography process, and then the insulating layer **104** and the etch stop layers **105** are partially removed through the etch process until the interconnect wirings **106** disposed between two adjacent MIM capacitor structures **100** are revealed in an opening **O4** and the sixth electrode layer **116c2** is revealed in an opening **O5**.

[0072] Referring FIG. **32**, the conductive material may fill up with the opening **O4** and the opening **O5** by a deposition process and/or a plating process (e.g., electroplating, electro-less plating, etc.). In various embodiments, the conductive material may include copper, aluminum, or the like. After forming the conductive material within the opening **O4** and the opening **O5**, a planarization process may be performed to remove excess of the conductive material from over the insulating layer **104** and to form a first metallic contact **208a**, second metallic contact **208b** and upper interconnect wirings **206**. The upper interconnect wirings **206** may be further electrical connected to an external bonding structure (e.g., a solder bump, a micro-bump, or the like) (not shown). The first electrode layer **112a**, the third electrode layer **112b**, the fifth electrode layer **112c1** are electrical connected to

the external bonding structure through the interconnect wirings **106**, the conductive vias **108**, the first metallic contact **208a** and the upper interconnect wirings **206**. The second electrode layer **116a**, the fourth electrode layer **116b** and the sixth electrode layer **116c2** are electrical connected to the external bonding structure through the second metallic contact **208b** and the upper interconnect wirings **206**.

[0073] FIG. **33** to FIG. **37** illustrate the cross-sectional views of intermediate stages in the formation of a MIM device in accordance with some embodiment.

[0074] Referring FIG. **33**, an insulating layer **122** is deposited on the capping layer **150** and the first dielectric layer **114a** of FIG. **28**. A nitride layer **124** is deposited on the insulating layer **122**. In some embodiments, the insulating layer **122** and the nitride layer **124** may be formed by the same processes and the same materials as the insulating layer **122** and the nitride layer **124**.

[0075] Referring FIG. **34**, an eighth spacer **148** in the opening O2 may be formed by the same processes and the same materials as the first spacer **132** after depositing the insulating layer **122** and the nitride layer **124**. A ninth spacer **140** in the opening O3 may be formed by the same processes and the same materials as the first spacer **132** after depositing the insulating layer **122** and the nitride layer **124**. The eighth spacer **148** and a ninth spacer **140** are formed during the same process. Unless further description as follows, the processes, the materials and the effects as of forming the eighth spacer **148** and the ninth spacer **140** are the same as the first spacer **132** of FIG. **3** to FIG. **4**, and will not be repeated herein.

[0076] The eighth spacer **148** is formed on a third portion of the first dielectric layer **114a** to cover the capping layer **150**. The ninth spacer **140** is formed on a portion of the fifth dielectric layer **114e** to cover a sidewall of the capping layer **150**, and a first sidewall S23 of the sixth electrode layer **116c2**. Then, the first dielectric layer **114a**, the first electrode layer **112a** and the fifth dielectric layer **114e** are patterned by using the eighth spacer **148**, the ninth spacer **140** and the capping layer **150** as a mask, to reveal a second sidewall S21 of the first dielectric layer **114a**, a second sidewall S22 of the first electrode layer **112a**, a first sidewall S24 of the fifth dielectric layer **114e**, and the insulating layer **104**. Then, MIM capacitor structures **200** are formed. A single MIM capacitor structure **200** is an asymmetrical structure. The adjacent MIM capacitor structures **200** may be mirror-symmetrical to each other.

[0077] Since the first dielectric layer **114a** and the first electrode layer **112a** are etched according to the eighth spacer **148**, the eighth spacer **148** has an outermost sidewall that is substantially aligned with the second sidewall S21 of the first dielectric layer **114a** and the second sidewall S22 of the first electrode layer **112a**.

[0078] Since the fifth dielectric layer **114e** is etched according to the ninth spacer **140**, the ninth spacer **140** has an outermost sidewall that is substantially aligned with the first sidewall S24 of the fifth dielectric layer **114e**.

[0079] In some embodiments, a height H8 of the eighth spacer **148** is greater than the height H7 of the seventh spacer **146**. An eighth lateral offset D8 is between the second sidewall S22 of the first electrode layer **112a** and the second sidewall S20 of the second electrode layer **116a**, and the eighth lateral offset D8 substantially equals to a maximum width W8 of the eighth spacer **148**. In some embodiments, the eighth spacer **148** is disposed over a lateral offset region of the first electrode layer **112a**.

[0080] In some embodiments, a height H9 of the ninth spacer **140** is less than the height H8 of the eighth spacer **148**. A sum of the height H9 of the ninth spacer **140** and the height H4 of the fourth spacer **138** is less than the height H8 of the eighth spacer **148**. A sum of the height H9 of the ninth spacer **140** and the height H7 of the seventh spacer **146** is less than the height H8 of the eighth spacer **148**. A ninth lateral offset D9 is between the first sidewall S23 of the sixth electrode layer **116c2** and the first sidewall S24 of the fifth dielectric layer **114e**, and the ninth lateral offset D9 substantially equals to a maximum width W9 of the ninth spacer **140**. In some embodiments, the ninth spacer **140** is disposed over a lateral offset region of the fifth dielectric layer **114e**.

[0081] Referring FIG. 35 to FIG. 37, except for the MIM capacitor structures **200**, the processes and the materials may be the same processes and the same materials as FIG. 30 to FIG. 32 and will not be repeated herein. In FIG. 37, the first metallic contact **208a**, second metallic contact **208b** and the upper interconnect wirings **206** are formed and may be further electrical connected to an external bonding structure (e.g., a solder bump, a micro-bump, or the like) (not shown). The first electrode layer **112a**, the third electrode layer **112b**, the fifth electrode layer **112c1** are electrical connected to the external bonding structure through the interconnect wirings **106**, the conductive vias **108**, the first metallic contact **208a** and the upper interconnect wirings **206**. The second electrode layer **116a**, the fourth electrode layer **116b** and the sixth electrode layer **116c2** are electrical connected to the external bonding structure through the second metallic contact **208b** and the upper interconnect wirings **206**.

[0082] A capacitor structure and a method for fabricating the capacitor are provided. The method for forming the capacitor structure including at least one pair of self-aligned spacers may decrease the number of the photomasks utilized in patterning processes of the capacitor, thereby lowering the fabrication costs of the capacitor. Accordingly, reliability and capacitance of the capacitor with the self-aligned spacers are also improved.

[0083] In accordance with some embodiments of the present disclosure, a method for fabricating a capacitor is provided. The method includes: forming a first electrode layer, a first dielectric layer covering the first electrode layer, a second electrode layer covering the first dielectric layer, a second dielectric layer covering the second electrode layer and a third electrode layer covering the second dielectric layer; patterning the third electrode layer to reveal the second dielectric layer and a first sidewall of the third electrode layer; forming a first spacer on a first portion of the second dielectric layer to cover the first sidewall of the third electrode layer; patterning the second dielectric layer and the second electrode layer by using the third electrode layer and the first spacer as a mask to reveal the first dielectric layer and a first sidewall of the second electrode layer; forming a second spacer on a first portion of the first dielectric layer to cover the first sidewall of the second electrode layer; patterning the first dielectric layer and the first electrode layer by using the third electrode layer, the first spacer and the second spacer as a mask to reveal a first sidewall of the first electrode layer; and forming a first conductive layer on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer. In some embodiments, the second spacer further covers a first sidewall of the second dielectric layer. In some embodiments, the first conductive layer is spaced apart from the second electrode layer by the first spacer and the second spacer. In some embodiments, the first conductive layer is in contact with a top surface of the third electrode layer and the first sidewall of the first electrode layer. In some embodiments, the method further includes: forming a first metallic contact electrically connected to an interconnect wiring; and forming a second metallic contact electrically connected to the second electrode layer. In some embodiments, the method further includes: after forming the first conductive layer, forming a third dielectric layer covering the third electrode layer; patterning the third electrode layer to reveal the second dielectric layer and a second sidewall of the third electrode layer; forming a third spacer on a second portion of the second dielectric layer to cover the second sidewall of the third electrode layer; patterning the second dielectric layer and the second electrode layer by using the third dielectric layer and the third spacer as a mask to reveal the first dielectric layer and a second sidewall of the second electrode layer; and forming a second conductive layer on the third spacer and the third dielectric layer, wherein the second conductive layer is electrically connected to the second electrode layer. In some embodiments, the method further includes: forming a trench in an insulating layer, wherein the first electrode layer, the first dielectric layer, the second electrode layer and the second dielectric layer are disposed in the trench and extend over a top surface of the insulating layer. In some embodiments, the insulating layer is formed over an interconnect wiring of an interconnect structure, and the first electrode layer lands on and is electrically connected to the interconnect wiring.

[0084] In accordance with some embodiments of the present disclosure, a capacitor structure is provided. The capacitor structure includes a first electrode layer, a first dielectric layer, a second electrode layer, a second dielectric layer, a third electrode layer, a first spacer, a second spacer, and a conductive layer. The first dielectric layer is disposed on the first electrode layer. The second electrode layer covers the first dielectric layer. The second dielectric layer covers the second electrode layer. The third electrode layer covers the second dielectric layer. The first spacer is disposed on a first portion of the second dielectric layer to cover a sidewall of the third electrode layer. The second spacer is disposed on a first portion of the first dielectric layer to cover a sidewall of the second electrode layer. The conductive layer is disposed on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer. In some embodiments, a thickness of the third electrode layer and substantially equals to a height of the first spacer. In some embodiments, a sum of a thickness of the second electrode layer, a thickness of the second dielectric layer and a thickness of the third electrode layer substantially equals to a height of the second spacer. In some embodiments, a height of the first spacer is less than a height of the second spacer. In some embodiments, the conductive layer is spaced apart from the second electrode layer by the first spacer and the second spacer. In some embodiments, the conductive layer is in contact with a top surface of the third electrode layer and a sidewall of the first electrode layer. In some embodiments, a first lateral offset is between the sidewall of the second electrode layer and the sidewall of the third electrode layer, and the first lateral offset substantially equals to a maximum width of the first spacer, and a second lateral offset is between the sidewall of the first electrode layer and the sidewall of the second electrode layer, and the second lateral offset substantially equals to a maximum width of the second spacer. In some embodiments, a lateral offset is between the sidewall of the second electrode layer and the sidewall of the third electrode layer, and the lateral offset substantially equals to a maximum width of the first spacer.

[0085] In accordance with some embodiments of the present disclosure, a capacitor structure is provided. The capacitor structure includes a first electrode layer, a second electrode layer, a third electrode layer, a first spacer, a second spacer, and a conductive layer. The first electrode layer, the second electrode layer and the third electrode layer are spaced apart from each other by dielectric layers. The first spacer is disposed over a first lateral offset region of the second electrode layer to cover a sidewall of the third electrode layer. The second spacer is disposed over a second lateral offset region of the first electrode layer to cover a sidewall of the second electrode layer. The conductive layer is disposed on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer. In some embodiments, a height of the first spacer is less than a height of the second spacer. In some embodiments, the conductive layer is in contact with a top surface of the third electrode layer and a sidewall of the first electrode layer. In some embodiments, the conductive layer is spaced apart from the second electrode layer by the first spacer and the second spacer.

[0086] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for fabricating a capacitor, comprising: forming a first electrode layer, a first dielectric layer covering the first electrode layer, a second electrode layer covering the first dielectric layer, a

second dielectric layer covering the second electrode layer and a third electrode layer covering the second dielectric layer; patterning the third electrode layer to reveal the second dielectric layer and a first sidewall of the third electrode layer; forming a first spacer on a first portion of the second dielectric layer to cover the first sidewall of the third electrode layer; patterning the second dielectric layer and the second electrode layer by using the third electrode layer and the first spacer as a mask to reveal the first dielectric layer and a first sidewall of the second electrode layer; forming a second spacer on a first portion of the first dielectric layer to cover the first sidewall of the second electrode layer; patterning the first dielectric layer and the first electrode layer by using the third electrode layer, the first spacer and the second spacer as a mask to reveal a first sidewall of the first electrode layer; and forming a first conductive layer on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer.

2. The method of claim 1, wherein the second spacer further covers a first sidewall of the second dielectric layer.

3. The method of claim 1, wherein the first conductive layer is spaced apart from the second electrode layer by the first spacer and the second spacer.

4. The method of claim 1, wherein the first conductive layer is in contact with a top surface of the third electrode layer and the first sidewall of the first electrode layer.

5. The method of claim 1 further comprising: forming a first metallic contact electrically connected to an interconnect wiring; and forming a second metallic contact electrically connected to the second electrode layer.

6. The method of claim 1 further comprising: after forming the first conductive layer, forming a third dielectric layer covering the third electrode layer; patterning the third electrode layer to reveal the second dielectric layer and a second sidewall of the third electrode layer; forming a third spacer on a second portion of the second dielectric layer to cover the second sidewall of the third electrode layer; patterning the second dielectric layer and the second electrode layer by using the third dielectric layer and the third spacer as a mask to reveal the first dielectric layer and a second sidewall of the second electrode layer; and forming a second conductive layer on the third spacer and the third dielectric layer, wherein the second conductive layer is electrically connected to the second electrode layer.

7. The method of claim 1 further comprising: forming a trench in an insulating layer, wherein the first electrode layer, the first dielectric layer, the second electrode layer and the second dielectric layer are disposed in the trench and extend over a top surface of the insulating layer.

8. The method of claim 7, wherein the insulating layer is formed over an interconnect wiring of an interconnect structure, and the first electrode layer lands on and is electrically connected to the interconnect wiring.

9. A capacitor structure, comprising: a first electrode layer; a first dielectric layer disposed on the first electrode layer; a second electrode layer covering the first dielectric layer; a second dielectric layer covering the second electrode layer; a third electrode layer covering the second dielectric layer; a first spacer disposed on a first portion of the second dielectric layer to cover a sidewall of the third electrode layer; a second spacer disposed on a first portion of the first dielectric layer to cover a sidewall of the second electrode layer; and a conductive layer disposed on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer.

10. The capacitor structure of claim 9, wherein a thickness of the third electrode layer and substantially equals to a height of the first spacer.

11. The capacitor structure of claim 9, wherein a sum of a thickness of the second electrode layer, a thickness of the second dielectric layer and a thickness of the third electrode layer substantially equals to a height of the second spacer.

12. The capacitor structure of claim 9, wherein a height of the first spacer is less than a height of the second spacer.

13. The capacitor structure of claim 9, wherein the conductive layer is spaced apart from the second

electrode layer by the first spacer and the second spacer.

14. The capacitor structure of claim 9, wherein the conductive layer is in contact with a top surface of the third electrode layer and a sidewall of the first electrode layer.

15. The capacitor structure of claim 14, wherein a first lateral offset is between the sidewall of the second electrode layer and the sidewall of the third electrode layer, and the first lateral offset substantially equals to a maximum width of the first spacer, and a second lateral offset is between the sidewall of the first electrode layer and the sidewall of the second electrode layer, and the second lateral offset substantially equals to a maximum width of the second spacer.

16. The capacitor structure of claim 9, wherein a lateral offset is between the sidewall of the second electrode layer and the sidewall of the third electrode layer, and the lateral offset substantially equals to a maximum width of the first spacer.

17. A capacitor structure, comprising: a first electrode layer; a second electrode layer; a third electrode layer, wherein the first electrode layer, the second electrode layer and the third electrode layer are spaced apart from each other by dielectric layers; a first spacer disposed over a first lateral offset region of the second electrode layer to cover a sidewall of the third electrode layer; a second spacer disposed over a second lateral offset region of the first electrode layer to cover a sidewall of the second electrode layer; and a conductive layer disposed on the first spacer and the second spacer to electrically connect the third electrode layer and the first electrode layer.

18. The capacitor structure of claim 17, wherein a height of the first spacer is less than a height of the second spacer.

19. The capacitor structure of claim 17, wherein the conductive layer is in contact with a top surface of the third electrode layer and a sidewall of the first electrode layer.

20. The capacitor structure of claim 17, wherein the conductive layer is spaced apart from the second electrode layer by the first spacer and the second spacer.
